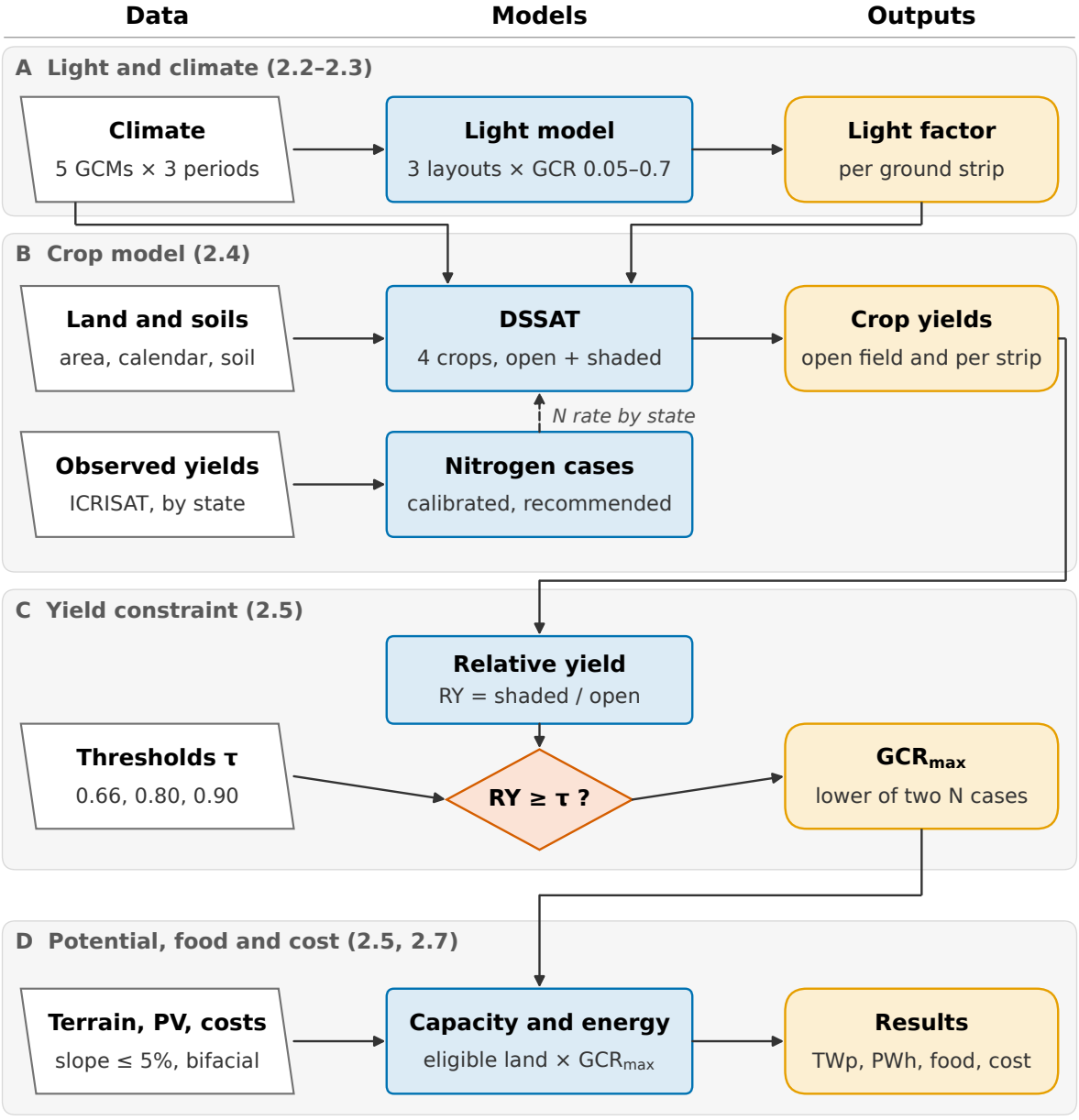
Data
Models
Outputs
A Light and climate (2.2–2.3)
Climate
5 GCMs × 3 periods
Light model
3 layouts × GCR 0.05–0.7
Light factor
per ground strip
B Crop model (2.4)
Land and soils
area, calendar, soil
DSSAT
4 crops, open + shaded
Crop yields
open field and per strip
N rate by state
Observed yields
ICRISAT, by state
Nitrogen cases
calibrated, recommended
C Yield constraint (2.5)
Relative yield
RY = shaded / open
Thresholds τ
0.66, 0.80, 0.90
RY ≥ τ ?
GCRmax
lower of two N cases
D Potential, food and cost (2.5, 2.7)
Terrain, PV, costs
slope ≤ 5%, bifacial
Capacity and energy
eligible land × GCRmax
Results
TWp, PWh, food, cost